Models	Acc (↑)	Acc@2 (↑)	% wins (↑)	κ (↑)
Human*	40.0	–	–	–
Random	33.3	66.6	–	–
Bottleneck Zero-shot Model. f: GPT-4, g: GPT-4, sequential				
Full model	39	77	–	–
w/o entailment	34 (-5)	71 (-6)	97	0.91
w/o ambiguity	35 (-4)	71 (-6)	90	0.91
w/o overconfidence	34 (-5)	71 (-6)	93	0.91
w/o half-truths	32 (**-7**)	68 (**-9**)	100	0.96
independent-full	36 (-3)	75 (-2)	93	0.85

Table 2: Performance of the models under ablation. Numbers in parenthesis denotes performance drop from the 'Full model'.

study that investigates the effect of each bottleneck control. The largest drop (7 points for Accuracy) occurs when the half-truth control is omitted, which aligns with our dataset analysis where real CC deliberately utters half-truths to deceive the judges, which, unless understood, may confuse the judges as well as models. We also find the quality of controls is better when derived sequentially (with conversation history) as compared to independent derivations, mirroring the original game setting.

Are few-shot models better than zero-shot models? Depends. For smaller or earlier LLMs (GPT-3, GPT-3.5), few-shot examples helps to improve their performance from zero-shot setup. However, for bottleneck models (and even in base/CoT models), GPT-4 achieves similar performance both in few-shot and zero-shot setups. Adding more demonstrations is often difficult due to LLM's limited context length. Further summarization (Park et al., 2023) or selection of few-shot examples (Madaan et al., 2022) are possible, but we leave it as a future work.

Are supervised models better than few/zero-shot models? No. Despite training on almost the full dataset (leave-one-out), the state-of-the-art supervised models do not outperform our zero-shot bottleneck models, indicating the superior ability of the LLMs to derive better bottleneck controls and act as a better discriminator.

Can models explain their chain-of-reasoning to detect deception? Yes. Table 1 shows results for human evaluation. In the pairwise comparison, our best model (bottleneck, f: GPT-4, g: GPT-4) wins unanimously against all other competing baselines. Indeed, the quality of the explanations is distinctly worse for bottleneck models using GPT-3 or GPT-3.5 when compared to GPT-4, denoting GPT-4's ability to better bottleneck controls, which form the basis of a good explanation. Similarly, base models cannot generate high-quality explanations mainly due to their unconstrained nature of generation. Figure 3b shows that our model has the highest e-ViL score, reflecting the trend from pairwise comparisons.

Do self-consistency improve reasoning performance? Not significantly. Wang et al. (2023) showed adding self-consistency prompt improves few-shot models reasoning performance. We implemented self-consistency in our best bottleneck model as well as in the COT baseline. Self-consistency improved the performance of the CoT baseline. However, it marginally improved the performance of our bottleneck models (not statistically significant). This indicates that despite using sophisticated task-specific prompts and refinement techniques, the dataset still remains a challenging case for state-of-the-art techniques involving LLMs. Table 3 shows the results of the self-consistency study for the CoT baseline and our bottleneck model (all with GPT-4).

Model	Acc (↑)	Acc@2 (↑)
CoT	32.0	64.7
CoT + Self-consistency	33.3 (+1.3)	68.0 (+3.3)
Ours	39.3 (+7.3)	77.3 (+12.6)
Ours + Self-consistency	40.0 (+8)	77.3 (+12.6)

Table 3: Performance of the models with self-consistency. Numbers in parenthesis denote performance increase from the 'CoT model.'

Are there any potential variations in performance across different language models? The choice of our language model (GPT-4) may significantly impact the results. A more comprehensive analysis of various models could strengthen the study. We extend our experiments by adding results from two more language models: Llama-2 (Touvron et al., 2023) and Bard[3]. Llama-2 is an open-source language model. While Bard (now Gemini) has access to Google Search and YouTube, the measures taken to prevent data leaks make Bard a fair candidate for a competing baseline. Table 4 shows the performance of GPT-4 dominating over other language models.

Qualitative analysis of the model generated explanations. Figure 4 exhibits model-generated

[3] https://gemini.google.com/

Model	Acc (↑)	Acc@2 (↑)
GPT-4 (2-shot)	**34.7**	**72.0**
Llama-2 (2-shot)	32.0 (-2.7)	64.7 (-7.3)
Bard (2-shot)	32.7 (-2.1)	65.3 (-6.7)
Bottleneck Models		
GPT-4 (*f*, *g*, 2-shot)	**39.3**	**77.3**
Llama-2 (*f*, *g*, 2-shot)	35.3 (-3)	71.3 (-6)
Bard (*f*, *g*, 2-shot)	36.0 (-3.3)	72.0 (-5.3)

Table 4: Performance across different language models.

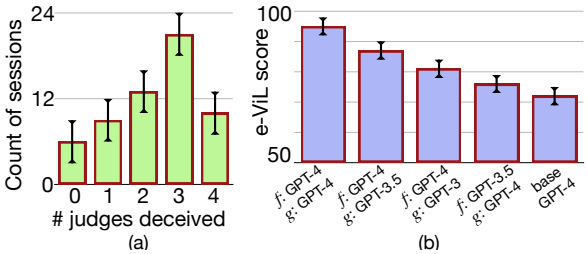

Figure 3: **(a)** Histogram of a number of judges who were deceived in session where our model predicted correctly. The distribution is skewed; skewness = -0.501, indicating our model predicts correctly, significantly ($p < 0.1$) more in sessions where more judges were deceived. **(b)** e-ViL scores for explanations from our top-5 accurate models, showing our best model generates more satisfactory explanations.

examples. In Figure 4a, the model correctly identifies the CC, whereas *All* of the judges fail. They mistook detailed information and deceptive appearances of the imposters (here, a swimmer's tan, athletic body, etc.). The model identifies bizarre details and randomness, even in the incorrect information by the imposters, while trying to capture the essence of the half-truths (highlighted) given by the real CC. In similar other cases where the model outperforms all the judges, the model mentions humor, overconfidence, deliberate incompleteness in answers, and wordy descriptions as possible cues of deceptive contestants.

In Figure 4b, the model predicts CC correctly, and the success for the judges is divided into half. Here, the model accurately recognizes a mistakenly erroneous response provided by the real CC as well as an "overall consistent behavior," denoting honesty. The judges who were not duped might have noticed CC's constant behavior or the inconsistencies among the imposters, while the duped judges might have relied on CC's inaccurate response to draw their conclusions. One of the judges who was not deceived in this case reappeared in the majority of the sessions, which indicates possible learning to support her claim (Banerjee et al., 2023).

Can models detect deception better than humans? Depends. Note that the human judges had access to all potential audio-visual and language cues in the original setting. Audio-visual cues are very important, when present, in deception detection primarily due to the ease and abundance of detecting them (DePaulo et al., 2003). Indeed, judges consistently use visual cues to explain their guesses, such as detecting the real swimmer/climber observing sun tan on the contestants or picking on the twinkle in the eyes for detecting a real joker. Considering this, our best model's performance is almost comparable with human performance—in absolute terms, our model incorrectly predicted 3/150 more examples compared to human judges. This is slightly different than results obtained in (Soldner et al., 2019), where they show linguistic features are the weakest predictors. However, it might be due to the fact that their linguistic features (part of them are LIWC features) were not expressive enough, or our dataset contains more linguistic cues than theirs. This also signifies the uniqueness of T4TEXT with the existence of the objective truth (affidavit), the basis of deception detection, and cross-questions revealing important factual and language cues to detect the real CC.

Do models detect deception for a complementary set than humans? Yes. Among the 59/150 correct predictions by our best model, we analyze how human judges performed in those game sessions. Figure 3a shows the histogram of the discrete variable–the number of judges deceived per session, considering only 59 sessions said above. The plot is skewed towards a higher number of deceived judges per session, indicating that our model performed better on a set where humans detect deception poorly. This implies our model discovered novel reasoning pathways, purely based on language cues, for correct predictions, which human judges failed at, even with the presence of multimodal cues and incentives. We qualitatively examine the explanations generated by the model where all judges got deceived (10/59) and discovered that our model generated more informative and plausible reasoning chains than explanations by human judges mentioned during the original sessions (see Figure 4).

(a) Model: Not deceived | Judges: All deceived | Correct answer: Number One

Number One: This contestant gave short, incomplete but true answers. For example, when asked about the boat she took to England, she said she didn't go by boat because she gets seasick. **This is a half-truth as she didn't specify how she actually got to England.** (...)
Number Two: This contestant gave long, detailed answers. (...) However, she was unsure about the name of the famous hotel in Montreal, which could indicate that she is an imposter.
Number Three: This contestant gave a random answer when asked about the time it took Gertrude Ederle to swim the English Channel. (...)

(b) Model: Not deceived | Judges: 2 deceived, 2 not deceived | Correct answer: Number One

Number One: This contestant gave generic and incomplete answers, which is a characteristic of the real person according to the hints. (..) They also struggled with the question about the size limit for a toy poodle, which could indicate unfamiliarity with the American measurement system, consistent with the affidavit stating that Maria is Brazilian. However, they failed to answer the question about candling an egg, which is a basic veterinary practice. **Despite their failure to answer one question, Number One's overall performance was most consistent.**
Number Two: This contestant gave **detailed and confident answers, which is a characteristic of an imposter.** They knew the correct (...) . However, they incorrectly stated (...).
Number Three: This contestant gave some correct answers, (...). However, they also gave some incorrect and vague answers. For example, they (...) suggested that the cure for a common chicken disease is to break the chicken's neck, **which is incorrect and unprofessional. Their answers are generally detailed and confident.**

(c) Model: Deceived | Judges: All not deceived | Correct answer: Number Three

Number One: He seems to be truthful and not overly detailed in his responses. **He doesn't know what a double clutch is**, which is a bit suspicious for a driving champion, but it's possible he never encountered it. His answer about the gas station address is generic but could be true. (...)
Number Two: He seems to be very detailed in his responses, which could indicate he's an imposter. (...) He also provides a specific address for his gas station, which could be made-up..
Number Three: He gives a random answer about parking near a hydrant, which could indicate he's making up facts. (...) His answer about double clutching is detailed and could be true, but it's also a bit overconfident. He provides a specific address for his gas station, which could be a made-up detail.
Based on these observations, I would **rank the contestants** in the following order of likelihood of not being an imposter: 1. Number One, **2. Number Three**, 3. Number Two.

Figure 4: Model generated explanations. Correct cues are colored in blue, and wrong cues are colored in red.

Can bottleneck features enhance human detection rate? **Yes**. We performed a correlation analysis between the presence of a bottleneck feature and human prediction jointly using a multivariate regression model on cases where having these features significantly helped the computational model to predict correctly. On the intersection where humans were wrong but the bottleneck model was correct, we find either no correlation ($p < 0.05$ for entailment and ambiguity) or negative correlation ($p < 0.05$ for overconfidence and half-truths), indicating humans when poor at detecting such linguistic cues suffers significantly in detecting lies.

Are some sessions more difficult than others? To the model, yes; to humans, mostly no. Banerjee et al. (2023) confirm no selection bias among the contestants across the sessions in the original game, indicating similar difficulty levels for the judges across sessions. They also highlight weakly significant selection bias among the judges, indicating better performance via learning and possible further selection as a judge in the session might have influenced their performance. Qualitatively, we see models' errors clustered around sessions where judges use non-language cues successfully; however, model predictions among themselves were highly correlated ($\rho = 0.7, p < 0.05$) for each underlying LLM variant. Figure 4c shows a case where the model fails to recognize the CC, but none of the judges were deceived. The model hallucinates in its reasoning to decide between Number One and Number Three as CC. It provides reasons for both to have incomplete, somewhat more detailed, a few factually accurate responses, and however forth—making the prediction at random. However, the real CC exists in the top two choices from the models, indicating judges picked up cues beyond language. Indeed, in this case, judges recognized the Southern accent of the CC and verified via the affidavit, a critical aural cue that was not available to our model.

6 Conclusion and Outlook

In this paper, we first showed the existence of a class of algorithmic detectors based on LLMs that can successfully identify language cues of deception without the presence of other visual or audio cues. We contribute a novel dataset T4TEXT for deception detection in the presence of objective truth and achieve a model performance comparable to human performance. We further find that our best model performs well in cases where humans perform poorly and discover novel language models that could augment human reasoning to detect deception, opening up the possibility of human-LLM collaborations to combat misinformation.

This paper advocates for human-AI collaboration, emphasizing the need for additional evidence on human dependence on algorithms in detecting textual deception. Using the methodology of incentivized decision-making from behavioral economics, our ongoing work seeks to observe and categorically elucidate human reliance on our AI model in different conditions (e.g., black box, glass box). Obtaining predictive features from our best model and those that drive human performance can provide us with a comprehensive understanding of why humans perform poorly in text-based deception detection tasks and whether AI tools can aid humans in such a crucial task that has lasting social, economic, and ecological impacts.